[Total No. of Question - 04 ]

[Total No. of Pages - 01 ]

# G H Raisoni International Skill Tech University, Pune

**F.Y B. Tech (TERM-II)**

**CAE I- Summer 2026 (2024 Pattern)**

## INTEGRAL CALCULUS AND DIFFERENTIAL EQUATIONS (25UBSL1203)

**(Time: - 01 Hour)** **(Max. Marks: 20)**

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**Instructions to the students:**

*i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.*
*iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed*
*v) Assume suitable data, if necessary.*

| Q.No. | *Question* | Marks | CO | BL |
|---|---|---|---|---|
| **Q1.**(a) | Trace the curve $x^2y^2 = a^2(y^2 - x^2)$. | [5] | 1 | 2 |
| b) | Prove that $\int_0^1 x^2(1-x)^3 dx = \frac{1}{60}$ | [5] | 1 | 3 |
| | **OR** | | | |
| **Q 2.**(a) | Find the value of $\int_0^\infty \sqrt{x}\, e^{-\sqrt{x}} dx$. | [5] | 1 | 2 |
| (b) | Find $\frac{dI(a)}{da}$, if $I(a) = \int_0^1 \frac{x^a - x^b}{\log x}\, dx$ | [5] | 1 | 2 |
| **Q 3.**(a) | Evaluate $\int_0^1 \int_0^x (3x^2 + y)\, dx\, dy$ | [5] | 1 | 2 |
| (b) | Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}$ dx dy by changing the order of integration | [5] | 1 | 2 |
| | **OR** | | | |
| **Q. 4**(a) | Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x\, dz\, dx\, dy$ | [5] | 1 | 2 |
| (b) | Calculate the volume of the solid bounded by the surface $x = 0, y = 0, x + y + z = 1$ and $z = 0$. | [5] | 1 | 3 |